

RADIP SHRESTHA

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PROFESSIONAL SUMMARY

Software Engineer with extensive experience developing AI-driven applications, real-time computer vision systems, and robust backend architectures. Specialized in building scalable Python/Flask services, optimizing PostgreSQL performance, and deploying secure Linux environments. Proven ability to bridge the gap between machine learning research and production-ready software solutions.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, PHP, C#, SQL, NoSQL

Frameworks & Tools: Flask, REST API Development, Git, Linux, Nextcloud, Bash, AWS

Databases: PostgreSQL, MongoDB

Data & Visualization: Pandas, NumPy, Matplotlib

System Administration: Ubuntu Server Setup, Networking, File Permissions, Privilege Management, Environment Inheritance

Certifications: Microsoft C# Certification (in progress)

PROJECTS

Real-Time Exercise Classifier & Form Analyzer (Python, TensorFlow, MediaPipe)

- Architected a multi-task Bi-LSTM + Temporal Attention model to simultaneously classify 7 exercise types and regress form-quality scores in real-time.
- Optimized inference latency using a sliding-window buffer, achieving 90%+ accuracy and sub-50ms feedback loops on standard consumer hardware.
- Integrated 33-point pose landmark extraction for high-fidelity motion tracking and kinematic analysis.

RAG Fitness Assistant (Python, Flask, LangChain, HuggingFace)

- Engineered a local RAG system utilizing hybrid BM25 and vector search (ChromaDB) to generate personalized workout plans from proprietary datasets.
- Developed custom LLM interceptors to enforce structured logic and state management, successfully reducing hallucination rates in multi-day workout generation.
- Automated data extraction from conversational chat to update SQLite-backed progress dashboards in real-time.

Full-Stack Library Management System (PHP, PostgreSQL)

- Designed a scalable backend featuring secure RBAC (Role-Based Access Control) for complex Librarian and Member workflows.
- Optimized PostgreSQL indexing and query execution plans, resulting in a significant reduction in search latency for high-volume catalog queries.
- Implemented comprehensive activity logging and session management to ensure 100% data auditability and integrity.

Supplement Store Location Recommendation System (Python, DBSCAN)

- Built a geospatial clustering model using DBSCAN to identify optimal supplement store locations based on gym density and competitor proximity.

- Processed and visualized spatial data using Pandas, NumPy, and Matplotlib.
- Generated density maps and cluster reports to support strategic business placement decisions.

Personal Cloud Server Deployment (Ubuntu, Nextcloud)

- Repurposed an old laptop into a secure, self-hosted cloud server using Nextcloud on Ubuntu.
- Configured remote access, user permissions, SSL, and automated backup routines.
- Documented the full deployment process for reproducibility and scalability.

EDUCATION

Northern Kentucky University | Bachelor of Science in Computer Science | **May 2026**

- Minor: Computer & Information Technology
- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Machine Learning, Operating Systems, Computer Networks, Deep Learning